Documenting Functions and Built-in Functions

Docstrings, annotations and help() explain your code; the built-ins are the toolbox you already own.

Docstrings

```
def bmi(weight_kg: float, height_m: float) -> float:
    """Return body-mass index.

    Args:
        weight_kg: weight in kilograms.
        height_m: height in metres.

    Returns:
        BMI rounded to one decimal place.
    """
    return round(weight_kg / height_m ** 2, 1)

print(bmi(70, 1.75))
print(bmi.__doc__.splitlines()[0])
print(list(bmi.__annotations__))
```

```
OUTPUT
22.9
Return body-mass index.
['weight_kg', 'height_m', 'return']
```

- The docstring is the **first statement** in the body; tools read it from `__doc__`.
- Line 1: short summary in the imperative mood ("Return..."), then a blank line, then details.
- Common layouts: **Google** (`Args:` / `Returns:` / `Raises:`), **NumPy** (`Parameters` + dashes), **reST** (`:param x:`).

Doctest: examples that double as tests

```
def double(n):
    """Return n doubled.

    >>> double(4)
    8
    """
    return n * 2

# check with:  python -m doctest -v file.py
```

Reading documentation

```
import inspect
print(inspect.signature(bmi))
print(bmi.__name__, bmi.__defaults__)
```

```
OUTPUT
(weight_kg: float, height_m: float) -> float
bmi None
```

Tool	Gives you
`help(fn)`	formatted docs in the console
`fn.__doc__`	the raw docstring
`fn.__annotations__`	type hints as a dict
`inspect.signature(fn)`	parameters + defaults
`dir(obj)`	names available on an object

Built-in functions you should know

Group	Functions
Numbers	`abs` `round` `pow` `divmod` `min` `max` `sum`
Conversion / types	`int` `float` `str` `bool` `list` `tuple` `set` `dict` `type` `isinstance`
Iteration	`range` `enumerate` `zip` `map` `filter` `reversed` `sorted` `iter` `next`
Truth tests	`any` `all` `callable` `bool`
I/O	`print` `input` `open`
Introspection	`len` `id` `dir` `vars` `help` `getattr` `hasattr`

```
nums = [4, -2, 9]
print(len(nums), sum(nums), min(nums), max(nums))
print(list(enumerate("ab", start=1)))
print(list(zip("ab", [1, 2])))
print(any(n < 0 for n in nums))
print(all(n < 10 for n in nums))
print(sorted(nums, reverse=True))
print(list(reversed(nums)), divmod(17, 5))
print(isinstance(3, int), round(3.14159, 2))
```

```
OUTPUT
3 11 -2 9
[(1, 'a'), (2, 'b')]
[('a', 1), ('b', 2)]
True
True
[9, 4, -2]
[9, -2, 4] (3, 2)
True 3.14
```

TIP `dir(builtins)` (after `import builtins`) lists every built-in; `help(sorted)` explains one.

PITFALL Don't shadow built-ins with your own names (`list`, `sum`, `max`, `id`, `input`) – later calls to the real function will break.